Alpha Forge Bitcoin Alpha Report

On-chain and macro ML modelling

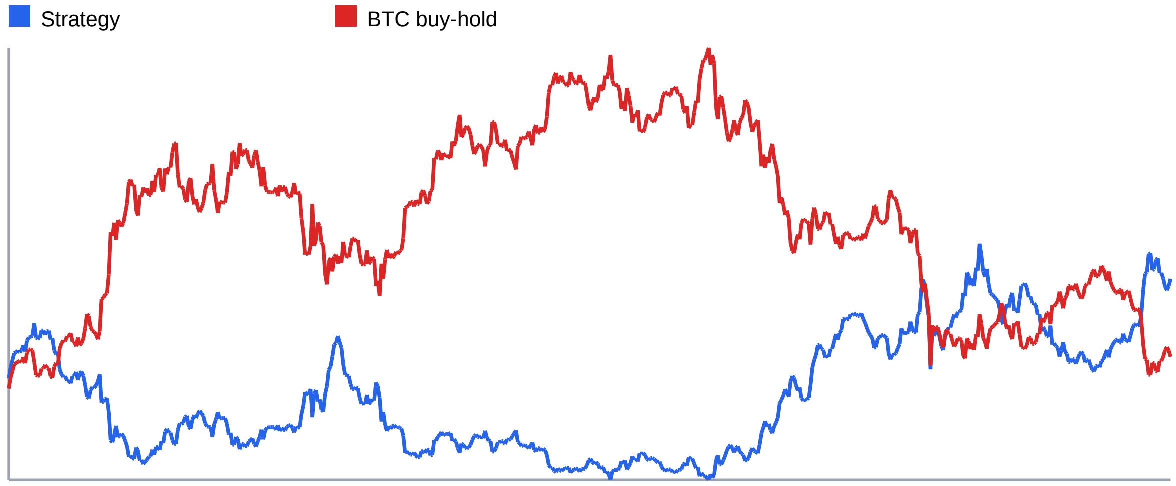
What I did

I cleaned the daily Bitcoin dataset, built the Rhode feature set, created a next-day log return target, trained several simple machine learning models, turned the best validation model into a daily long/short signal, and tested it against Bitcoin buy-and-hold. The split is chronological, so later rows never leak into earlier training.

Item	Value
Raw rows	4,292
Usable model rows	4,261
Model date range	2014-10-17 to 2026-06-16
Train	2014-10-17 to 2022-12-15
Validation	2022-12-16 to 2024-09-14
Test	2024-09-15 to 2026-06-16

Main result

The selected model was **Ridge alpha 80**, chosen on the validation set by the best strategy Sharpe ratio. On the untouched test period, it reached directional accuracy of 50.16%, an annualized Sharpe ratio of 0.60, and cumulative strategy return of 34.54%. Bitcoin buy-and-hold over the same test rows returned 8.81%.



Test metrics

Model	MAE	RMSE	Dir.	R2	Sharpe	Max DD	Strategy	BTC
Mean baseline	0.01679	0.02378	49.8%	-0.002	0.33	-51.2%	8.8%	8.8%
Boosted stumps 40	0.01699	0.02396	50.2%	-0.017	0.33	-51.2%	8.8%	8.8%
Boosted stumps 80	0.01733	0.02424	50.2%	-0.042	0.33	-51.2%	8.8%	8.8%
Boosted stumps 120	0.01739	0.02429	50.2%	-0.046	0.33	-51.2%	8.8%	8.8%
Ridge alpha 80	0.01969	0.02620	50.2%	-0.216	0.60	-43.1%	34.5%	8.8%
Ridge alpha 20	0.02027	0.02673	50.2%	-0.266	0.33	-51.2%	8.8%	8.8%
Ridge alpha 5	0.02046	0.02690	50.2%	-0.283	0.33	-51.2%	8.8%	8.8%
Ridge alpha 1	0.02052	0.02695	50.2%	-0.288	0.33	-51.2%	8.8%	8.8%

Feature checks

Feature	Correlation with next-day return
10Y Treasury yield	-0.0227
Active addresses	0.0227
Fee to market cap	0.0216
True Market Cap	-0.0203
Rolling volatility	0.0147
Hash rate	-0.0134
Fees in USD	0.0076
US Dollar Index	-0.0040

The raw correlations are weak, which is normal for daily Bitcoin returns. This means the model should be judged more by out-of-sample performance than by any single correlation value.

Market regimes

Regime	Days	Avg next return	Vol.	DXY	TNX	Active addr.
Bear	757	0.022%	3.82%	97.51	2.64	697,625
Bull	1,331	0.293%	4.06%	97.64	2.50	820,171
Consolidation	1,881	0.109%	2.90%	98.73	2.86	704,132
Macro shock	202	-0.194%	2.63%	102.76	2.95	754,903

Feature importance

Feature	Model weight	RMSE increase when shuffled
Active addresses	0.373	0.000112
True Market Cap	0.337	0.000044
Rolling volatility	0.065	0.000006
Fee to market cap	0.016	-0.000000
Hash rate	0.022	-0.000002
10Y Treasury yield	0.070	-0.000002
US Dollar Index	0.039	-0.000007
Fees in USD	0.078	-0.000018

The strongest feature by permutation testing was **Active addresses**. This does not mean it predicts every day by itself. It means the trained model lost the most accuracy when that input was disturbed.

Data preparation notes

I used the eight input Rhode features and built 30-day rolling volatility from past log returns. Large on-chain values were log-transformed, then all model inputs were standardized using only the training period. The target was next-day log return. This lets the signal be used after today's close without seeing tomorrow's price. The long/short cutoff was calibrated on the validation period, then held fixed for the test period.

Conclusion

The signal is useful as a research alpha, not as a finished trading system. It shows that on-chain activity, network valuation, macro conditions, and volatility contain some information about the next day's return direction. The main weakness is that daily Bitcoin returns are noisy, so small statistical gains can disappear after trading costs, slippage, or regime changes. A production version should add transaction costs, position sizing, walk-forward retraining, and stricter risk limits.